An Adaptive and Link-Based Method for Video Scene Clustering and Visualization

Hong Lu, Kai Chen, Yingbin Zheng, Zhuohong Cai, and Xiangyang Xue

Abstract In this paper we propose to adaptively cluster video shots into scenes on page rank manner and visualize video content on clustered scenes. The clustering method has been compared with state-of-arts methods and experimental results demonstrate the effectiveness of the proposed method. In visualization, the importance of the shots in the scene can be obtained and incorporated into the visualization parameters. The visualization results of the test videos are shown in global and detail manner.

Keywords Video scene clustering • Page-rank • Visualization

1 Introduction

As video content are becoming more and more distributed, easily obtained, and edited, it is necessary to present the video content in a more compact and informative manner for browsing, navigation, retrieval, edition, etc. In [1], a radial tool is presented to hierarchically visualize structured video content. Specifically, a system which combines automatic video segmentation and a characteristic key-frame extraction is proposed. In the outer circular rings, the segmented video are shown. And in each segment, logos are detected and shown in the rings. The duration of the logo appearance is mapped to the circle segment size.

In [2], *Jesen–Shannon divergence (JSD)* is employed to measure the difference between neighboring video frames to segment a video into shots, and also choose key frames in each shot. Then an innovative 3D visualization approach is proposed. In the visualization, video key frames and the useful information related to the

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process of key frame selection are displayed in different layers of detail. The visualization tool can help for a quick and clear understanding to the video content.

In [3], volume visualization techniques are used for summarizing video sequences and presenting the video content. Min Chen et al. [4] propose visual signature for symbolizing abstract visual features that depict individual objects and motion events.

Hangzai Luo et al. [5, 6] analyze the news web and propose to incorporate the knowledge of news video for video visualization. It can provide more informative information on the news videos and is useful for users to browse, navigate and understand the news video content. Shiliang Zhang et al. [7] focus on music videos and propose not only to perform the personalized MV affective analysis, but also novel affective visualization to convert the abstract affective states intuitively and friendly to users.

Thus, how to effectively analyze the video content and extract interesting, important information, and also with detail tuning to present video content to users is important for video analysis, browse, and retrieval.

The remainder of the paper is organized as follows. Section 2.1 reviews video scene clustering and pagerank based methods. And pagerank clustering method is presented in Sect. 2.2. The video visualization method is presented in Sect. 3. Scene clustering and visualization results are given in Sect. 4. And we conclude our work in Sect. 5.

2 Video Scene Clustering on PageRank

2.1 Video Scene Clustering and PageRank Method

In content based video analysis, the first step is shot boundary detection, which can be regarded as well solved problem. Then on the detected shots, how to group shots of a video into clusters of disparate scenes and extract compact information is a key step toward high-level video semantic understanding, event identification, indexing, and retrieval [8]. Much work has been proposed for video scene segmentation and summarization [9–15]. These methods can be grouped into that based on normal methods such as spectral method, Markov Chain Monto Carlo (MCMC) [13, 15], and that based on graph partitioning methods [10–12, 14].

Since graph connectivity can well represent that two shots are similar, much work has been proposed to model scene segmentation as a graph partitioning problem. Specifically, in [10], audio and visual features are extracted first for each shot. Then scene transition graph [11] is constructed by considering the visual and audio feature similarity and time constraint. The time constraint means that if two shots are far in temporal order, although the visual and audio similarities are high, the final similarity between two shots is low. In [12], a video coherence

measure is proposed by adopting the normalized cuts criterion that emphasizes the homogeneity between shots in the same scene and inhomogeneity between those in different scenes. A graph partitioning method is proposed in [14] for video scene segmentation.

Based on the constructed graph, link based analysis can be performed. These years, link based analysis has been widely used in region-of-interest detection, image search, and event recognition [16–19]. In this paper, we present an efficient approach to cluster video shots into scenes based on link analysis.

It is well known that Google uses pagerank [20] to rate the webs in the Internet and makes great success. In these years, pagerank has been widely accepted and applied in other fields. For example, in [16], inter-image level and intra-image level link analysis is performed to find the highly ranked images as exemplars of objects and to localize object instances in each image. In [17], pagerank is used for large-scale image search. Yushi Jing and Shumeet Baluja [18] and Naveed Imran et al. [19] successfully use pagerank to rank the pictures and informative features in pictures that finish picture search and event recognition. The theme lies that the most informative pictures and features are those which have more links to other pictures and features.

Due to the success of using pagerank in computer vision tasks, we propose to use pagerank for video scene clustering and visualization. First, it can rank the node values in the set well and can be used for clustering. Second, the representative shots in the cluster can be obtained at the same time for video scene visualization and summarization. The detail description will be in Sect. 2.2.

2.2 PageRank Clustering

In this section, we present our proposed method for video scene clustering by using pagerank. First, the video is segmented into a sequence of shots $\mathcal{S} = \{S_1, S_2, \ldots, S_M\}$, say M of them, by using a popular shot-boundary detector [21, 22]. Each shot is considered as a basic unit for clustering. The *shot color histogram* (SCH)—the bin-wise median of all the frame color histograms within the shot [23]—is computed as the shot feature for scene clustering. The frame, and thus the shot, color histogram is obtained in HSV (Hue-Saturation-Value) color space with its color coordinates uniformly quantized into 12 (Hue), 4 (Saturation), and 4 (Value) bins, respectively, resulting in a total of 192 bins (or quantized colors). The color space is selected due to its good performance in other work of image and video analysis [24, 25]. Then, the similarity between two shots are measured by intersection on histograms of two shots, i.e. $\mathcal{S}(s_i, s_j) = \sum_{k=1}^{N} min(X_i(k), X_j(k))$.

Same to that in [14], the scene likeness is defined based on the shot similarity.

$$\mathcal{L}(s_i, s_j) = \begin{cases} 1, \mathcal{S}(s_i, s_j) \geq \tau \\ 0, otherwise \end{cases} \tag{1}$$

Since pagerank is fitted to digraph and the scene likeness can be reformed into digraph, we make the connection bidirectional. The pagerank value is computed as below.

$$PR(u) = \sum_{v \in B_u} \frac{PR(v)}{L(v)} \tag{2}$$

where the PR value for a node u is dependent on the PR values for each node v out of the set B_u (it contains all nodes linking to node u), divided by the number $L(v)$ of links from node v. And the algorithm is described in Algorithm 1.

In the algorithm, the parameters α is used to control the compactness of the clusters. If α is large, the formed clusters are more compact. Based on our empirical study, it works well when α lies in range of $\left[\frac{1}{5*|V_k|}, \frac{1}{|V_k|}\right]$.

Algorithm 1 PageRank clustering algorithm

Definitions:
V: Set of nodes;
C: Set of shots under consideration for inclusion in V;
R: Set of shots that have not been assigned to any scene clusters.
Algorithm:

```
k ← 0
R = V
While R ≠ ∅
    k ← k + 1
    s_1^k = arg max_{s_i ∈ R} L(s_i, R − {s_i})
    V_k ← s_1^k
    R ← R − {s_1^k}
    C_k ← {s_i | L(s_i, s_1^k) = 1, s_i ∈ R}
    R ← R − C_k
    j ← 1
    While C_k ≠ ∅
        for s_i ∈ C_k
            PR = PageRank(V_k ∪ s_i)
            if PR(s_i) ≥ α
                j ← j + 1
                s_j^k = s_i
                V_k ← V_k ∪ {s_j^k}
                C_k = C_k − {s_i}
                C_k ← C_k ∪ {s_i | L(s_i, s_j^k) = 1, s_i ∈ R}
                R ← R − {s_i | L(s_i, s_j^k) = 1, s_i ∈ R}
            else
                R = R ∪ {s_i}
                C_k = C_k − {s_i}
```

3 Video Content Visualization

During video scene clustering, we can also represent the video content in a graph using the clustering parameters. Specifically, for each shot, a key frame is extract from it. Then the clustered shots are represented with links between them in the graph. The size of the frame in each shot is proportional to the pagerank value as computed in Eq. (2) of the corresponding shot. In other words, the larger the pagerank value of the shot, it is shown larger in the visualized scene cluster graph. And the not clustered shots, i.e. each shot is a cluster, are shown separately in the graph.

Figure 1 shows the scene clustering result by using pagerank for Tennis1 video. In the figure, the formed clusters are the wide-angle views of the court, close-up views of the players, medium views of the players and the court, and other clusters containing only one or two shots of transition scenes or commercial breaks,

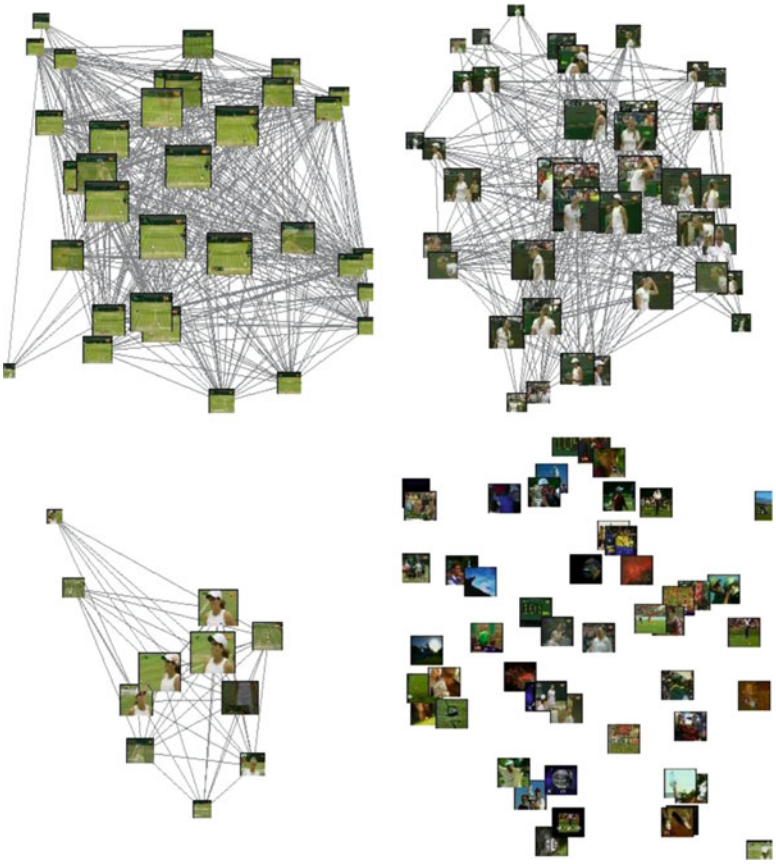

Fig. 1 Scene clustering result by using pagerank for Tennis1 video

respectively. From the figure, we can easily extract the informative, representative shots of the video, i.e., the clusters formed, and the bigger images in each scene can represent the representative content of the cluster.

4 Experiments

4.1 Experimental Data

The experimental data used is tabulated in Table 1. The video data include sports and news videos. And the videos are digitized at 25 frames/s and the durations range from 7 to 47 min. To measure the clustering performance, we manually label the scene type of each shot as the ground truth.

4.2 Performance Comparison

We compare our proposed link-based video scene clustering method with that by using k-means and normalized cuts methods. For the k-means method, we use the code in Matlab 7.1. And the normalized cuts (Ncuts) code is from the website.[1] For k-means and Ncuts, the cluster number is determined by the Davies–Bouldin index (DBI). The DBI is a cluster validity measure that has been widely used, shown to be effective [26], and employed in some existing studies of video clustering [27].

In our experiments, we have chosen to evaluate the quality of a clustering result by using the Minkowski measure [28]. This measure is computed based on a cluster co-membership matrix C, with its entry $C_{i,j}$ equal to 1 or 0 depending on whether or not shots S_i and S_j belong to the same scene cluster. Denote the co-membership matrix of the desired clustering result as T and that of a clustering result obtained as C, respectively. The Minkowski measure of the clustering result C is defined as the normalized norm distance between the two matrices, i.e. $D(T,C) = \frac{||T-C||_\gamma}{||T||_\gamma}$, where γ is the selected norm order. When $\gamma = 1$, the Minkowski measure can be expressed as $D(T,C) = D_m + D_f$, where $D_m = \frac{\sum_{i,j} \mathbf{1}(T_{i,j}=1 \& C_{i,j}=0)}{\sum_{i,j}|T_{i,j}|}$ and $D_f = \frac{\sum_{i,j} \mathbf{1}(T_{i,j}=0 \& C_{i,j}=1)}{\sum_{i,j}|T_{i,j}|}$.

Table 1 Experimental data

Video data	Tns1	Tns2	Bkt1	Bkt2	Vlb1	Vlb2	Soc1	Soc2	News1	News2
Frame #	17981	24171	22014	24192	31608	11664	32301	12562	71378	57462
Shot #	137	141	194	195	240	56	179	48	476	342
Scene #	4	3	4	4	4	3	3	2	5	5

[1] http://www.cis.upenn.edu/~jshi/software/.

Table 2 Performance comparison

	k-means				Normalized cuts				Pagerank			
Video	K_{KM}	D_{KM}	D_m	D_f	K_{NC}	D_{NC}	D_m	D_f	K_{PR}	D_{PR}	D_m	D_f
Tns1	7	1.26	0.44	0.82	54	0.66	0.62	0.04	56	0.58	0.49	0.09
Tns2	8	0.78	0.32	0.46	45	0.75	0.69	0.05	38	0.63	0.56	0.06
Bkt1	10	0.97	0.55	0.42	65	0.80	0.71	0.09	56	0.88	0.63	0.26
Bkt2	6	1.42	0.23	1.19	35	0.78	0.65	0.13	71	1.04	0.76	0.29
Vlb1	9	0.63	0.41	0.22	68	0.64	0.62	0.02	67	0.92	0.62	0.30
Vlb2	5	1.62	0.09	1.53	40	0.54	0.53	0.01	15	0.67	0.63	0.04
Soc1	8	0.98	0.63	0.35	50	0.83	0.79	0.04	17	1.03	0.42	0.61
Soc2	2	0.60	0	0.60	27	0.71	0.71	0.01	4	0.16	0.12	0.04
News1	20	4.21	0.25	3.96	18	0.92	0.90	0.02	209	0.87	0.86	0.01
News2	18	8.57	0.03	8.54	26	0.92	0.91	0.01	210	0.89	0.88	0.01
Mean		2.10	0.30	1.81		0.76	0.71	0.04		0.77	0.60	0.17

Table 3 Computational complexity of scene clustering methods

Method	Complexity
k-means	$O(KnMT) + O(nM^2)$
Normalized cuts	$O(nMT) + O(nM^2)$
Pagerank clustering	$O(nM^2)$

D_m and D_f denote the miss and false classification rates, respectively. Consequently, the lower the Minkowski measure, the smaller the miss and false classification rates, and the better the scene clustering result.

The experimental results on the performance comparison are shown in Table 2. In the table, K_{KM}, K_{NC}, and K_{PR} represent the cluster numbers determined by using the k-means, Ncuts, and pagerank methods. In pagerank based method, τ is 0.6 and α is $\frac{1}{3.5*|V_k|}$.

It can be observed from Table 2 that our method can obtain better clustering performance than k-means and comparable performance with normalized cuts method.Also, our method can obtain more small clusters.

4.3 Complexity Analysis of Video Scene Clustering Methods

In this section, we analyze the computational complexity of the methods and tabulated in Table 3. In k-means and normalized cuts methods, the maximum cluster number is set to $\sqrt{M}$, where M is the number of shots. And DBI measure is computed based on the clustering results to determined the optimal cluster number and to get the corresponding clustering result.

Let K be the number of the resulting scene clusters, M the total number of shots, n the feature dimension, and T the number of k-means iterations before convergence. The computational complexity of k-means clustering can be shown to be O

Table 4 Computing times (in seconds) taken by k-means clustering (T_{KM}), normalized cuts (T_{NC}), and the proposed pagerank (T_{PR}) clustering method

Video data	Tns1	Tns2	Bkt1	Bkt2	Vlb1	Vlb2	Soc1	Soc2	News1	News2
T_{KM}	109.99	114.88	361.20	347.52	733.25	8.93	261	5.42	6265.60	2330.83
T_{NC}	40.37	44.27	116.52	133.41	218.63	2.77	98.41	2.07	1980.01	678.79
T_{PR}	2.77	1.57	2.65	3.12	3.63	0.76	1.83	0.40	9.10	8.57

(*KnMT*) [29]. Considering to compute DBI for determining the optimal cluster number, an additional complexity of $O(nM^2)$ is required by k-means. Similar analysis is done to normalized cuts method.

In our experiment, we used a Intel Core 2 Duo 2.53GHz PC with 2G memory. The time comparison is shown in Table 4. It can be observed from Table 4, since our proposed pagerank based method process a binary scene likeness matrix and not on the original high dimensional shot features, our method performs much faster. The improvement is much more when the video has much larger number of shots or the resulted number of clusters.

4.4 Video Visualization

Based on the clustering results, we can visualize the video on the clusters, the representative shots in the scene, the link between the shots. Figure 2 shows the visualization results for News1 video. In the figure, we can observe the news story part to some talking show, some soccer match, etc., and the anchor person scene.

On the other hand, Fig. 3 gives the accumulated time coverage information of the first five formed clusters in the test videos. The time coverage is computed on the number of frames of the formed scenes to the total number of frames of the video. It can be observed from Fig. 3 that in sports videos, the first formed clusters can have large time coverage of the whole video and can be termed as *dominant scenes*. And in news videos, the first formed scenes are anchor person view and the news story scenes having high visual similarity. Since the news stories are normally diverse, the accumulated time coverage of the first formed scenes is not much high.

5 Conclusions

We propose in this paper of video shot clustering based on pagerank, which considers the similarity of video shots as link between shots. In the clustered scenes, the shots within that which have larger links with other shots are visualized larger. The parameters can be obtained during shot clustering. Experimental results demonstrate that the clustering method can obtain better performance and can also provides a compact and informative content.

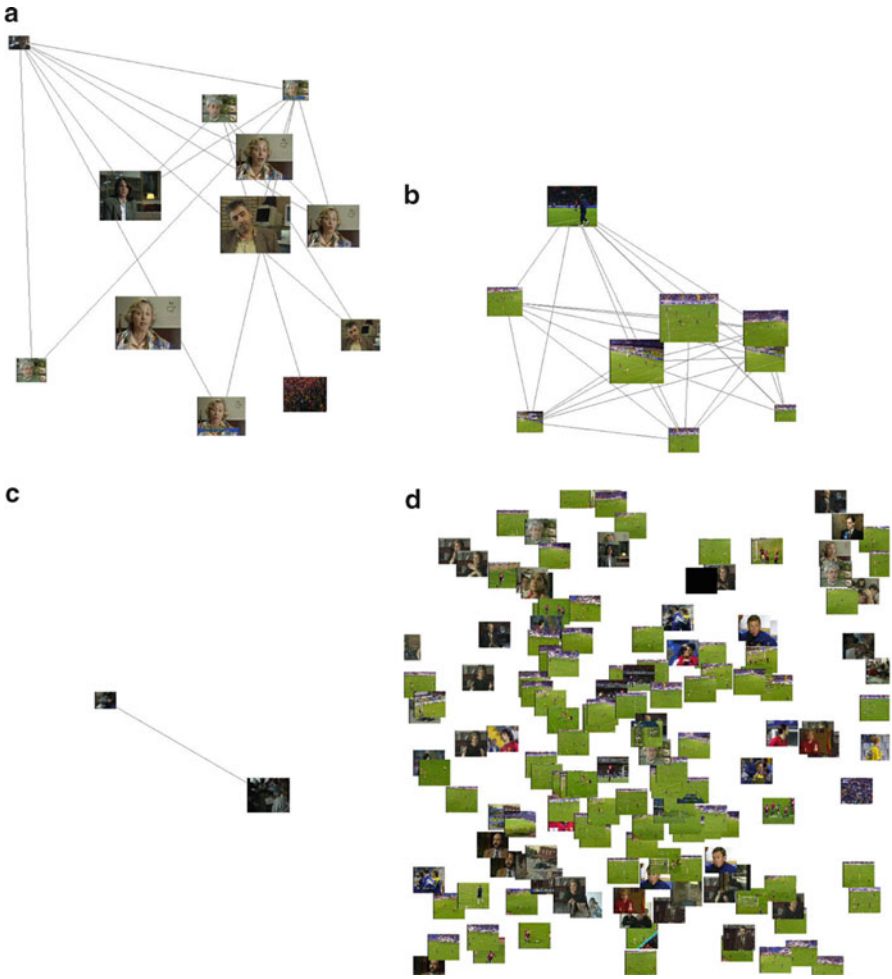

Fig. 2 Scene clustering result by using pagerank for News1 video

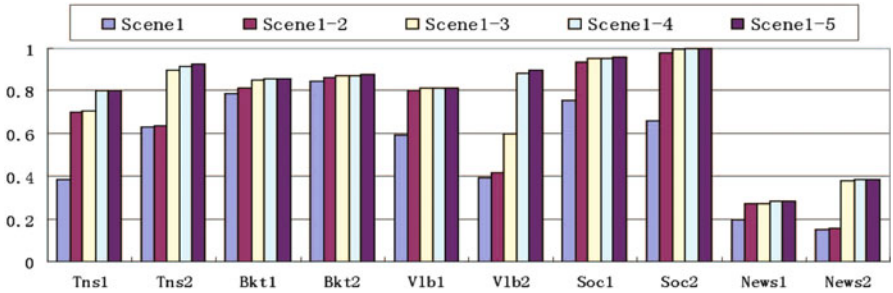

Fig. 3 Time coverage information of the first five formed scenes

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